

POWER MANAGEMENT STUDY



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Department of Mechanical Engineering
Faculty of Engineering
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Team Members



Prof. W.P.D. Fernando
Project Supervisor



H.M.W. Gunawardana
Undergraduate
Researcher



H.A.T. Sanvidu
Undergraduate
Researcher



M.N.D. Fernando
Undergraduate
Researcher



N.N.A. Kumarasiri
Undergraduate
Researcher



A.P.I. Udayanga
Undergraduate
Researcher



P.M.S.S. Wijethunga
Undergraduate
Researcher



K.S. Nethkalana
Undergraduate
Researcher

Abstract

Sri Lanka's power sector faces critical challenges due to escalating electricity demand and heavy reliance on expensive, carbon-intensive thermal generation during peak hours. While solar photovoltaic energy capacity is expanding rapidly, its intermittent nature presents substantial grid integration challenges without bulk energy storage. Concurrently, Sri Lanka possesses an extensive existing hydroelectric infrastructure with untapped potential for conversion into Pumped Storage Hydropower (PSH) systems. This independent research study proposes an integrated power management and planning framework designed to optimize renewable energy utilization across the national grid. The proposed approach investigates leveraging surplus daytime solar generation to pump water from lower to upper reservoirs, effectively transforming conventional hydro infrastructure into grid-scale "solar batteries." Furthermore, the study incorporates demand forecasting and resource assessment across hydro, solar, and waste-to-energy assets into a centralized planning framework. The expected contribution is a comprehensive strategic roadmap and decision-support framework that provides a sustainable foundation for long-term power generation planning in Sri Lanka, aiming to minimize thermal generation reliance and reduce operational costs.

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1. Introduction

1.1 Background

Sri Lanka's electricity sector is undergoing a crucial transition driven by industrial growth and increasing domestic requirements. Traditionally dependent on large-scale hydroelectricity and imported fossil fuels, the national grid faces immense strain during evening peak demand. To mitigate environmental impacts and volatile fuel import costs, national energy policies heavily emphasize expanding renewable energy, particularly solar. However, the variable nature of solar generation necessitates highly efficient energy management and bulk storage solutions to ensure a stable grid.

1.2 Problem Statement

The primary challenge in the current power generation mix is the misalignment between renewable availability and peak demand. Renewable energy sources, such as solar, are intermittent and reach maximum output during off-peak daytime hours. Consequently, expensive thermal power plants are heavily relied upon to stabilize the grid during evening peaks. While existing hydroelectric resources present significant, untapped potential for large-scale energy storage, there is an urgent need for a comprehensive planning approach that better synchronizes and utilizes these renewable resources to displace thermal generation.

1.3 Aim of the Study

The overall aim of this study is to formulate a comprehensive power management and planning framework that evaluates the integration of surplus daytime solar energy with pumped storage hydroelectricity to reduce dependency on thermal power and enhance grid sustainability.

1.4 Objectives

General Objective:

- To develop a holistic power plant management framework and planning roadmap for optimizing renewable energy integration and bulk storage.

Specific Objectives:

- Study the current power generation infrastructure across the national grid.
- Investigate the potential of upgrading existing facilities to pumped storage hydropower.
- Identify suitable locations and reservoir configurations for future developments.
- Study future electricity demand trends and historical usage patterns.

- Explore strategies for integrating daytime renewable energy to pump and store water.
- Develop a centralized planning framework for sustainable power management.

1.5 Scope of the Study

The scope of this research encompasses:

- Existing hydro power plants and reservoir configurations.
- Solar energy utilization and ground coverage constraints.
- Pumped storage opportunities within the current infrastructure.
- Power demand studies and future trend modeling.
- Renewable energy planning and dispatch allocation.
- Preliminary optimization concepts for generation management.

Note: This study focuses exclusively on high-level system planning, analytical modeling, and feasibility analysis. It does not include detailed civil/mechanical engineering design, structural blueprints, or hardware implementation.

2. Proposed Study

2.1 Study Overview

This study proposes a multi-layered analytical approach that connects existing generation assets, renewable expansion, demand forecasting, and storage mechanisms into a centralized framework. The structural roadmap of the proposed study is illustrated in Figure 2.1.

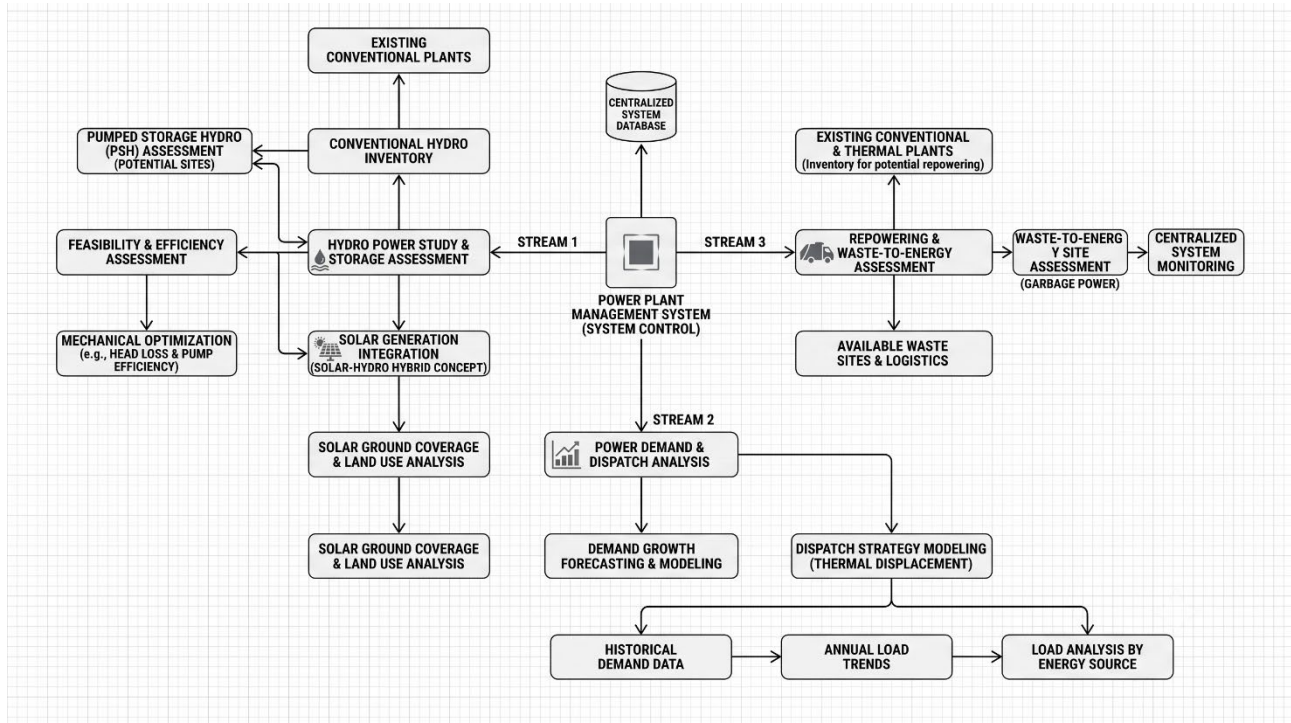


FIGURE 2.1: POWER MANAGEMENT STUDY ROADMAP

2.2 Existing Power Generation

The study will examine the baseline capabilities of Sri Lanka's current power generation fleet. This includes reviewing the capacities and operational constraints of existing hydro power plants, existing thermal plants, and current renewable energy sources (including waste-to-energy and solar installations).

2.3 Hydro Power Assessment

A core focus of the study is investigating existing hydro plants and their associated reservoir configurations. The assessment will evaluate the geometric and mechanical potential for pumped storage retrofitting, alongside identifying geographic opportunities for future developments.

2.4 Solar Energy Integration

The study will examine the national solar energy potential, specifically focusing on daytime surplus generation. It will detail how excess solar energy, which often exceeds daytime grid requirements can be utilized directly for water pumping, effectively charging the hydro-reservoirs.

2.5 Power Demand Study

Understanding load dynamics is essential for effective grid management. The study will review historical demand patterns, investigate future demand trends, and explore data-driven demand forecasting approaches to determine accurate storage capacity requirements.

2.6 Power Management Strategies

Different operational approaches will be explored to improve the utilization of renewable energy. By simulating various pumping and generation schedules, the study aims to establish strategies that systematically reduce dependence on thermal generation during peak hours.

2.7 Waste-to-Energy Assessment and Integration

While solar and hydro resources form the primary focus for peak-shifting and bulk storage, municipal solid waste-to-energy (WTE) plants offer a critical, consistent baseload alternative. The study will conduct a strategic assessment of existing garbage power plants and map available logistical sites for potential future developments. By integrating these WTE facilities into the centralized monitoring system, the framework will evaluate how steady generation from waste resources can effectively supplement the intermittent nature of solar power, further minimizing the need to dispatch conventional fossil-fuel thermal plants.

2.8 Centralized Planning Framework

The study will synthesize information from the resource assessments, demand predictions, and management strategies into a proposed centralized framework. This framework is designed to integrate different study components to support high-level decision-making and long-term planning for grid operators.

3. Study Methodology

The study will follow a structured, high-level analytical methodology:

- Literature review of renewable integration and pumped storage.
- Data collection from national generation logs.

- Study of existing power infrastructure and operational limits.
- Site investigation for candidate reservoir geometries.
- Demand analysis to baseline current load profiles.
- Renewable energy assessment for solar availability.
- Framework development integrating all system parameters.
- Evaluation and recommendations for strategic implementation.

4. Expected Outcomes

The study aims to produce the following deliverables:

- An assessment of existing renewable power resources across the grid.
- Identification of potential pumped-storage opportunities within the hydro network.
- A centralized framework for renewable power planning and management.
- Actionable recommendations for improving renewable energy utilization.
- A robust, data-informed foundation for future detailed technical and engineering studies.

5. Project Schedule

Figure 2.1 illustrates the established project roadmap, which has been structured to prioritize sequential data acquisition and analytical modeling across the study's primary streams. The project team operates on a consistent, weekly operational meeting schedule to ensure progress tracking, coordinate resource allocation across the distinct work groups, and conduct iterative technical reviews of findings.

The study has successfully initiated its first phase, prioritizing foundational data gathering and preliminary feasibility screenings. Current active work streams are focused on centralizing power demand logs for long-term growth modeling, initiating comprehensive operational inventories of existing conventional hydro and waste-to-energy assets, and establishing the key technical parameters for the detailed feasibility and efficiency assessments.

6. Conclusion

Transitioning toward a sustainable grid requires innovative planning and efficient energy storage. This study addresses the urgent need to offset expensive thermal generation by proposing an integrated framework that harnesses surplus solar energy to power pumped storage hydroelectricity. By providing a clear strategy for managing these resources, the study's expected contribution will serve as a vital stepping stone for sustainable, long-term power planning in Sri Lanka.

References

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